

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 01
Course Title : Advanced Programming Language
Course Code : cse 342
Experiment Name:

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FOR Loop:

```
public class ForLoopExample {  
    public static void main(String[] args) {  
  
        System.out.println("Using FOR loop:");  
  
        for (int i = 1; i <= 5; i++) {  
            System.out.print(i + " ");  
        }  
    }  
}
```

Output:1 2 3 4 5

WHILE Loop:

```
public class WhileLoopExample {  
    public static void main(String[] args) {  
  
        System.out.println("Using WHILE loop:");  
  
        int i = 1;  
  
        while (i <= 5) {  
            System.out.print(i + " ");  
            i++;  
        }  
    }  
}
```

Output:1 2 3 4 5

DO-WHILE Loop:

```
public class DoWhileLoopExample {  
    public static void main(String[] args) {  
  
        System.out.println("Using DO-WHILE loop:");  
  
        int i = 1;  
  
        do {  
            System.out.print(i + " ");  
            i++;  
        } while (i <= 5);  
    }  
}
```

Output:1 2 3 4 5

Discussion:

In this experiment, we implemented three types of loops in Java: for loop, while loop, and do-while loop. All three loops were used to print numbers from 1 to 5. We observed that the output of all loops is the same, but their execution process is different. The for loop is mainly used when the number of iterations is known in advance. The while loop checks the condition before executing the loop body, so it may not run if the condition is false. On the other hand, the do-while loop executes the loop body first and then checks the condition, so it always runs at least one time. This makes do-while different from the other two loops. Through this experiment, we clearly understood how loop structures work in Java.